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RESEARCH ARTICLE

Toward Interpretable Hybrid AI: Integrating Knowledge Graphs and Symbolic Reasoning in Medicine

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ABSTRACT Knowledge Graphs (KGs) are data structures that enable the integration of heterogeneous data sources and supporting both knowledge representation and formal reasoning. This paper introduces TrustKG, a KG-based framework designed to enhance the interpretability and reliability of hybrid AI systems in healthcare. Positioned within the context of lung cancer, TrustKG supports link prediction, which uncovers hidden relationships within medical data, and counterfactual prediction, which explores alternative scenarios to understand causal factors. These tasks are addressed through two specialized hybrid AI systems, **VISE** and **HealthCareAI**, which combine symbolic reasoning with inductive learning over KGs to provide interpretable AI solutions for clinical decision-making. Leveraging KGs to represent biomedical properties and relationships, and augmenting them with learned patterns through symbolic reasoning, our hybrid approach produces models that are both accurate and transparent. This interpretability is particularly important in medical applications, where trust and reliability in AI-driven predictions are paramount. The empirical analysis demonstrates the effectiveness of **VISE** and **HealthCareAI** in improving the predictive accuracy and clarity of model outputs. By addressing challenges in link prediction—such as discovering previously unknown connections between medical entities—and in counterfactual prediction, TrustKG, with **VISE** and **HealthCareAI**, underscores the potential of integrating KGs with symbolic AI to create trustworthy, interpretable AI systems in healthcare. This paper contributes to the advancement of semantic AI, offering a pathway for robust and reliable AI solutions in clinical settings.

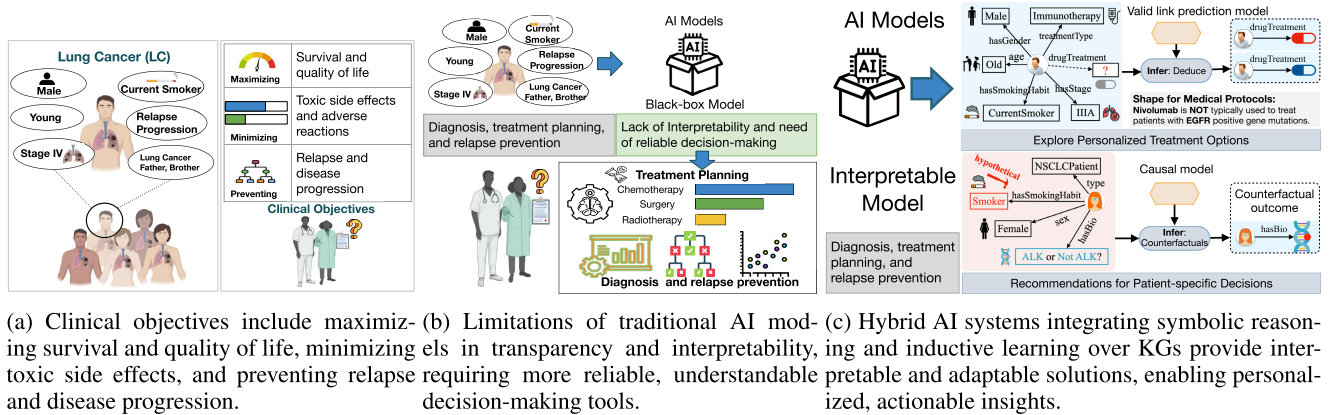
INDEX TERMS Counterfactual prediction, inductive learning, knowledge graphs, link prediction, symbolic learning.

I. INTRODUCTION

Knowledge graphs (KGs) provide a robust framework for integrating data and metadata from diverse sources [1]. By bridging disparate data silos and creating an integrated view, KGs enable powerful insights in complex domains

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that require comprehensive data analysis, such as healthcare. In lung cancer, for instance, diagnosis and treatment decisions are intricate, with oncologists needing to consider numerous factors—including tumor type, stage, and genomic markers [2]—to develop patient-specific treatment plans. These factors guide crucial decisions regarding therapies, dosages, and treatment cycles, which are tailored to individual patient responses and comorbidities [3], [4]. The primary clinical



(a) Clinical objectives include maximizing survival and quality of life, minimizing toxic side effects, and preventing relapse requiring more reliable, understandable and disease progression. (b) Limitations of traditional AI models in transparency and interpretability, enabling personalized, actionable insights. (c) Hybrid AI systems integrating symbolic reasoning and inductive learning over KGs provide interpretable and adaptable solutions, enabling personalized, actionable insights.

FIGURE 1. AI Models in Medicine. Figures (a) and (b) illustrate clinical objectives for LC treatment and the limitations in transparency of traditional AI models. Hybrid AI integrating symbolic reasoning with inductive learning over KGs presents a promising approach to overcoming these challenges, combining interpretability and adaptability to offer clinicians deeper tailored to specific patient needs (Figure (c)).

objectives in lung cancer management include: i) *Maximizing* survival and quality of life, ii) *Minimizing* side effects, and iii) *Preventing* relapse and disease progression [5].

While oncological decisions are guided by clinical expertise and established guidelines, the variability in patient cases often makes it difficult to determine optimal treatment strategies through intuition alone. Predictive models could assist in areas such as diagnosis, treatment planning, and relapse prevention [2], [3], [4], [6]. However, traditional models frequently lack interpretability and domain-specific insights, which are crucial in lung cancer management where precision and transparency are essential. Figures 1a and 1b illustrate the clinical objectives for lung cancer treatment and the limitations in the transparency of traditional AI models in the healthcare domain. To address these challenges, leveraging medical semantics can significantly enhance both the accuracy and interpretability of predictive models [7]. In particular, understanding *relapse risks* and *exploring personalized treatments* are essential for effective patient care in lung cancer [8]. These tasks can be approached through:

- **Relapse Risk Prediction through Valid Link Prediction:** Identifying latent connections between patient features (e.g., biomarkers) and relapse events helps to anticipate risk factors that may not be obvious in traditional datasets. This supports oncologists in preemptively adjusting treatment plans to mitigate relapse risks.
- **Personalized Treatment through Counterfactual Prediction:** Counterfactual reasoning allows the exploration of alternative treatment scenarios by simulating potential outcomes. This “what-if” analysis aids in identifying suitable treatment adjustments for individual patients, balancing relapse prevention and quality of life.

KGs capture complex clinical relationships, enabling valuable insights into tasks such as *relapse prediction* and *counterfactual prediction*, thereby contributing to improved patient care. Hybrid AI systems that integrate symbolic reasoning with inductive learning over KGs offer a promising

approach to these challenges. By combining the interpretability of symbolic methods with the adaptability of inductive learning, hybrid AI can deliver deeper, more actionable insights tailored to specific patient needs (Figure 1c). This paper presents a methodology for integrating knowledge graphs (KGs) with inductive learning, which facilitates the development of interpretable, context-aware artificial intelligence (AI) in the healthcare domain.

A. PROPOSED SOLUTION: TRUSTKG

The proposed solution, TrustKG, aims to address these challenges and deliver reliable, interpretable results to clinical users. TrustKG is a KG-based framework that supports semantic data management and KG analytics to promote interpretability in AI-driven healthcare. TrustKG facilitates tasks across data ingestion, processing, and integration, incorporating semantic alignment, named entity recognition, and entity linking/disambiguation. It includes tools for KG creation, validation, and exploration, enabling the representation of complex biomedical data with ontologies and logical reasoning, as well as validation through shape schemas. TrustKG’s analytics capabilities leverage both symbolic and numerical learning techniques for predictive modeling, including valid link prediction, counterfactual prediction, and causal analysis. This hybrid paradigm combines the strengths of both approaches: providing context-aware and consistent insights through symbolic reasoning and domain knowledge, while enhancing predictive accuracy by incorporating these insights into numerical learning models.

Specifically, the hybrid AI systems **VISE** and **HealthCareAI** are integrated within TrustKG, combining symbolic reasoning with inductive learning over KGs to address the problems of valid link prediction and counterfactual prediction, respectively. **VISE** uncovers latent relationships in the data to assess relapse risks, while **HealthCareAI** uses counterfactual reasoning to explore personalized treatment options. The valid connections discovered by **VISE** provide a foundation for **HealthCareAI**, improving the accuracy

and contextual relevance of its counterfactual predictions. Together, these systems enhance transparency and reliability in clinical decision-making by integrating interpretability with predictive precision. TrustKG also emphasizes human-centered communication by providing a natural language interface and referencing scientific publications to support its recommendations, empowering clinicians to make informed, evidence-based decisions.

B. CONTRIBUTIONS AND STRUCTURE

Building on previous work on **WISE** [9] and **HealthCareAI** [10], this paper makes the following novel contributions:

- The development of TrustKG, an interoperable and interpretable knowledge graph (KG) ecosystem tailored for the medical domain.
- The formalization of the problems of *Valid Link Prediction* and *Counterfactual Prediction* as critical tasks for improving accuracy and interpretability in clinical decision-making systems.
- The specification of **WISE** and **HealthCareAI** as complementary hybrid AI systems, where **WISE** uncovers valid connections in the data to address *Valid Link Prediction*, providing a foundation for **HealthCareAI** to enhance the accuracy and contextual relevance of *Counterfactual Prediction* through symbolic reasoning combined with inductive learning.
- Empirical evaluation of **WISE** and **HealthCareAI** on KGs created from a lung cancer dataset, demonstrates the impact of semantics on predictive performance.

The paper is structured as follows: Section II reviews related work, while Section III describes the main features implemented in TrustKG. Section IV defines the problem of valid link prediction and presents **WISE**, along with the results of an experimental evaluation of its performance. Similarly, Section V introduces the problem of counterfactual prediction and positions **HealthCareAI** as a hybrid AI system that leverages semantics to effectively solve this task. Section VI examines the limitations of the methods implemented in TrustKG and highlights the challenges that remain unresolved. Finally, Section VII provides concluding insights and future directions.

II. RELATED WORK

A. KGS IN MEDICINE

In the field of medicine, Knowledge Graphs (KGs) have been widely employed to address challenges in semantic data integration through approaches like knowledge extraction [3] and exploration [11]. Chandak et al. [12] introduce PrimeKG, a KG that unifies clinical concepts from diverse medical sources, encompassing 17,080 diseases with alignments to biological processes, experimental drugs, and protein alterations. PrimeKG enables forecasting of drug-disease interactions and treatment recommendations. Sakor et al. [13] present Knowledge4COVID-19, a framework for constructing healthcare KGs using rule-based entity

linking and machine learning. This KG facilitates tasks such as drug-drug interaction prediction, treatment recommendations for COVID-19, and treatment impact visualization. Knowledge4COVID-19 showcases the potential of structured clinical knowledge extraction from unstructured data. In oncology, Aisopos et al. [3] propose DE4LungCancer, a health data ecosystem leveraging controlled vocabularies and ontologies to represent lung cancer patients' medical histories. The ecosystem uses the RML mapping engine [14] for KG creation from diverse data sources, and SHACL technologies for data quality validation. Trav-SHACL [15], an efficient SHACL validation engine, assesses entities (e.g., patients) against specific medical protocols. Ristoski and Paulheim [16] explore KG-based data linking and integration, supporting effective data discovery and knowledge exploration. Rivas and Vidal [17] propose an inductive learning approach using graph neural networks to detect drug-drug and drug-target interactions, predicting molecular properties and potential treatments. Callahan et al. [18] introduce an open-source KG ecosystem for life sciences, presenting a methodology to construct large-scale KGs with components for data pre-processing, KG construction, and analytics. In lung cancer research, Virginia et al. [2] develop the TTR KG, which integrates knowledge on non-small cell lung cancer, including data from 12,351 patients with attributes such as smoking habits and mortality rates. The TTR KG provides analytical insights and supports survival rate predictions. Empirical evaluation of **WISE** and **HealthCareAI** on KGs created from a lung cancer dataset, demonstrating the impact of semantic integration on predictive performance.

B. AI MODELS IN MEDICINE

Recent advancements in AI have brought personalized healthcare systems closer to reality. Janik et al. [19] propose a personalized care model for lung cancer patients to estimate relapse probability, comparing the accuracy of traditional AI models (78%) with graph machine learning models (68%) for relapse prediction. Pan et al. [20] investigate the effectiveness of AI models, such as Support Vector Machine and Random Forest, in predicting relapse for acute lymphoblastic leukemia patients. Similarly, Yang et al. [21] explore how machine learning techniques, including decision trees and deep neural networks, can analyze the influence of clinical status and demographics on the survivability of early-stage cancer patients. Additional studies [6], [22] examine AI techniques for predicting lung cancer mutations from tabular and image data, highlighting the impact of attributes like prescribed treatments on predictive accuracy. One closely related study explores the use of machine learning models, particularly random forests [23], in personalized healthcare. Vyas et al. [24] model patient-level and patient-episode health records, developing ensemble-based predictive models for dementia prognosis and personalized treatment recommendations, with interpretability provided by LIME [25]. This approach helps oncologists forecast and recommend treatments, enhancing decision-making in

clinical settings. While AI models show promise for predictive tasks in medicine and oncology, such approaches often lack interpretability and do not fully leverage domain-specific knowledge, limiting applicability in clinical settings. Integrating semantic information through KGs offers a pathway to address these limitations, enhancing interpretability and enabling AI systems to provide more context-aware insights for personalized patient care.

C. EXPLAINABLE AI IN MEDICINE

As AI adoption in medicine grows, a key challenge persists: many models function as black boxes, and the decision processes remain opaque and difficult to interpret. Understanding the rationale behind AI predictions is essential for trust and reliability, especially in healthcare. Suh et al. [26] highlight the communication gap between AI practitioners and medical experts, showing that accuracy metrics alone are insufficient for building understanding among subject matter experts. Instead, models must be explicitly interpretable to provide meaningful insights. To address this need, Li et al. [27] identify distinct user personas in Knowledge Graph (KG) applications—KG Builder, KG Analyst, and KG Consumer—each with unique requirements for interpretability. Similarly, Purohit and Vidal [28] propose the *DIGGER* pipeline, which extracts logical rules from lung cancer treatment data to reveal patterns, flag protocol deviations, and complete missing relationships within KGs, providing clinicians with transparent, explainable insights. Neuro-symbolic AI approaches have furthered explainability by combining symbolic and numerical methods. For instance, Rivas et al. [29] predict therapy efficacy using a hybrid learning approach, while Chudasama et al. [30], [31] introduce InterpretME, a KG-based framework that traces and explains predictive model outcomes with LIME and SHAP [32]. In the SemDesLC framework [33], Semantic Web technologies are leveraged to make lung cancer relapse predictions interpretable, bridging gaps between medical insights and AI predictions. Despite these advancements, limitations remain. Current models often lack domain-specific semantics, which restricts the capture of complex medical context essential for clinical decision-making. Many systems also struggle to integrate heterogeneous data effectively, limiting the interpretability of complex relationships. These challenges highlight the need for frameworks like TrustKG, which integrates symbolic reasoning with inductive learning over KGs, enabling semantically enriched, context-aware insights that support clinical decisions in complex healthcare scenarios.

D. AI MODELS FOR CAUSAL REASONING

Advanced predictive models have been widely used in medicine for diagnosis support, prognosis, and personalized treatment recommendations [34], [35], [36]. However, while these models provide valuable predictions, limitations exist in terms of interpretability and trustworthiness—key

elements in clinical decision-making where understanding the “why” and “what if” behind recommendations is crucial [37], [38]. Causal models, which address these questions, require assumptions about dependencies among clinical variables, such as the Stable Unit Treatment Value Assumption (SUTVA) and the Ignorability Assumption [39]. To enhance causal analysis, Salimi et al. [40] developed CaRL, a framework for causal reasoning over relational databases, while Huang et al. proposed CareKG and CauseKG to model causal relationships and estimate causal effects in KGs [41], [42]. However, these models fall short in handling counterfactual prediction (“what if” scenarios), which requires comprehensive causal graphs (CGs) to capture all relationships. Techniques like structural causal models (SCMs) and causal Bayesian networks (CBNs) allow counterfactual predictions when a CG is available, but CG construction is challenging and often relies on domain experts or data-driven methods like the Peter-Clark (PC) algorithm and Greedy Equivalence Search (GES) [43]. Despite recent advances, generating complete causal graphs from data remains theoretically unattainable [37]. Emerging methods using large language models for causal discovery still lack integration of clinical semantics, which could improve CG accuracy. Additionally, Dickerman et al. [44] explore KG embeddings for counterfactual prediction without a CG, with real-world effectiveness yet to be validated. Moreover, existing causal models in healthcare lack comprehensive semantic integration, which limits the interpretability of the outcomes. These gaps are addressed through the integration of semantic reasoning with KGs, enabling richer causal insights and reliable counterfactual predictions tailored for complex clinical decision-making.

III. KGs AND AI MODELS IN MEDICINE

The application of AI in medicine, particularly in complex fields like oncology, requires systems that not only provide accurate predictions but also meet high standards of interpretability, reliability, and personalization. KGs offer a powerful approach for integrating heterogeneous medical data and encoding biomedical knowledge, making them ideal for use in hybrid AI systems. This section discusses the essential requirements for AI in medical applications, the structure and capabilities of TrustKG—a hybrid AI framework based on KGs—and introduces two key predictive tasks, valid link prediction and counterfactual prediction, relevant to improving decision-making in lung cancer treatment.

A. REQUIREMENTS FOR AI IN MEDICINE

The implementation of effective AI models to address predictive problems in medicine requires the satisfaction of several key requirements [3], [33], [45]:

- *Interpretability*: Oncologists need to understand the rationale behind AI-driven recommendations. This is crucial for validating model suggestions and making adjustments based on individual patient needs [46].

- **Reliability:** An AI model should provide consistent, accurate predictions, reducing the risk of erroneous recommendations that could negatively impact patient outcomes [47].
- **Context-Aware Reasoning:** AI models must incorporate medical knowledge, such as relationships between patient characteristics, tumor markers, and treatment outcomes. This knowledge enables AI systems to make decisions that align with clinical guidelines and patient-specific factors [48].
- **Personalization:** Effective AI models must be able to personalize treatment recommendations and account for patient-specific factors, such as genetic mutations, medical history, and potential comorbidities [3].
- **Adaptability:** AI systems should adapt to new medical knowledge and evolving treatment protocols. This requires a flexible framework capable of integrating updated data and knowledge sources over time [49].

Meeting these requirements is challenging with traditional machine learning approaches, due to inherent limitations in transparency and domain-specific knowledge [50]. To address these limitations, by combining semantics inferred from symbolic reasoning and integrity constraint validation with inductive learning over KGs.

B. TRUSTKG: A HYBRID AI FRAMEWORK FOR MEDICAL KNOWLEDGE GRAPHS

In the medical domain, the complexity of patient data requires AI systems that can integrate diverse, domain-specific knowledge while maintaining interpretability, reliability, and adaptability. Knowledge Graphs (KGs) are particularly suited to this challenge. A KG is a directed, edge-labeled graph that can be represented as $KG = (V, E, L)$ [51], where V represents nodes (e.g., entities like patients, treatments, or biomarkers), L denotes edge labels (e.g., relationships such as “has_symptom” or “treated_with”), and E is the set of edges connecting nodes based on these relationships. KGs enable the representation of medical knowledge, capturing complex interrelationships essential for accurate and context-aware AI-driven insights. In order to satisfy the criteria for interpretability, reliability, context-aware reasoning, personalization, and adaptability in clinical applications, a hybrid AI framework, dubbed TrustKG, is hereby proposed. This framework utilizes knowledge graphs (KGs) to facilitate sophisticated predictive tasks in the healthcare domain. TrustKG integrates symbolic reasoning with inductive learning, thereby yielding clinically pertinent insights. As illustrated in TrustKG is organized into four core layers—Semantic Data Management, Knowledge Graph Management, Knowledge Graph Analytics, and Human-Machine Communication (Figures 2a and 2b)—that together address challenges of medical data integration, knowledge management, interpretation, and usability.

Semantic Data Management facilitates the ingestion, processing, and integration of diverse data sources, including

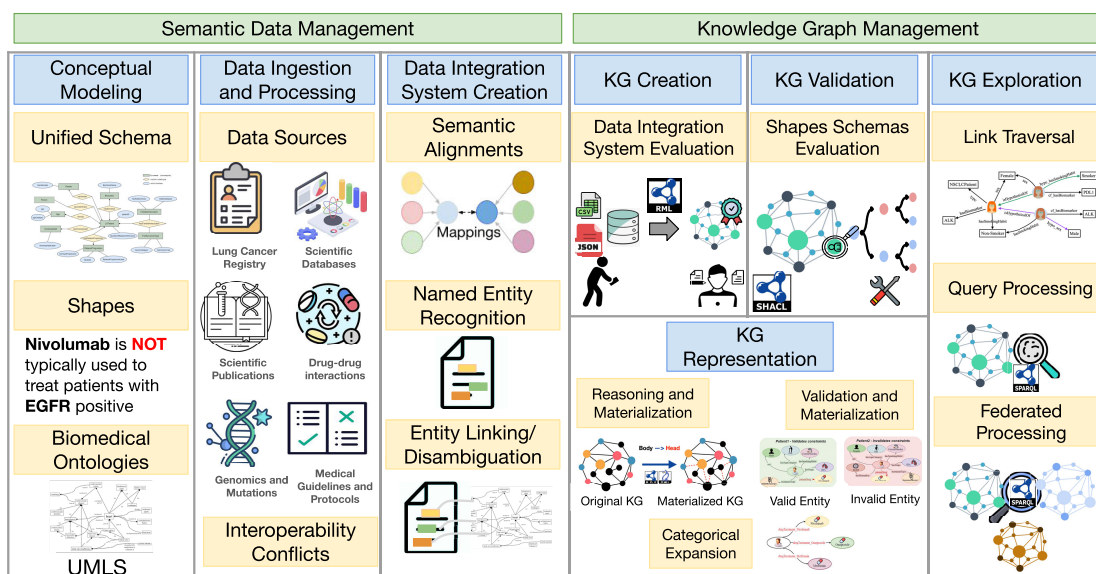
scientific literature, clinical guidelines, and patient records. This layer achieves data harmonization through semantic alignment, named entity recognition, and entity linking, creating a unified and interoperable schema that captures complex biomedical knowledge within an integrated data system. This structured approach enables the mapping of data sources to unified schemas or biomedical ontologies (e.g., the Unified Medical Language System - UMLS¹), while clinical protocols and guidelines are encoded as shape constraints. It is of paramount importance that domain-specific representations are employed to guarantee that the outputs produced by TrustKG are both interpretable and contextually relevant.

Knowledge Graph Management supports the creation, validation, and exploration of KGs, enabling reasoning and materialization. It uses shape schemas to validate the KG, ensuring the consistency and reliability of represented entities and relationships, and offering clinicians trustworthy insights. Federated query processing allows seamless exploration across multiple datasets, enhancing the system’s adaptability and usability in healthcare environments.

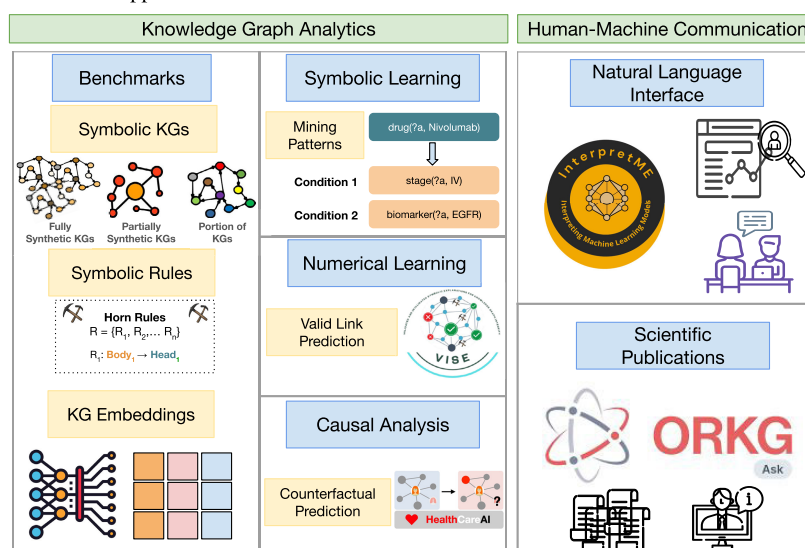
Knowledge Graph Analytics combines symbolic and numerical learning to enable predictive modeling. Symbolic learning identifies patterns within the graph, while numerical learning supports tasks such as valid link prediction and causal analysis. These capabilities directly address the need for context-aware reasoning by drawing insights from established medical knowledge and real-world data. In lung cancer treatment, for example, these analytics support tasks such as assessing risk of relapse and exploring treatment scenarios. In addition, various techniques will be implemented to generate benchmarks of synthetic KGs (i.e., fully and partially synthetic KGs [52]) for the empirical studies of TrustKG tools.

Human-Machine Communication provides a natural language interface and links insights to scientific publications, enhancing interpretability and trust. This layer ensures that clinicians receive recommendations in accessible language, allowing them to validate insights against existing literature and make informed decisions. By presenting results transparently, TrustKG fosters clinician confidence in the AI’s output. Additionally, TrustKG leverages frameworks such as InterpretME [30], a KG-driven system that enables fine-grained representation of the main characteristics of trained machine learning models, ensuring that clinicians and researchers can better understand model behaviors and predictions. TrustKG also integrates tools like ORKG Ask [53], a hybrid AI system that supported on vector search, large language models, and KGs allows for the scholarly search and exploration. In TrustKG, ORKG Ask provides advanced query and exploration capabilities, facilitating access to relevant scientific knowledge and enhancing clinicians’ ability to interpret AI-driven insights in a broader research landscape.

¹ Accessed on November 14th, 2024: <https://www.nlm.nih.gov/research/umls/quickstart.html>



(a) **Semantic Data Management and Knowledge Graph Management in TrustKG.** The Semantic Data Management layer supports conceptual modeling, data ingestion, processing, and integration across heterogeneous data sources, utilizing techniques like semantic alignment, named entity recognition, and entity linking to harmonize medical data. The Knowledge Graph Management layer focuses on creating, validating, and exploring KGs, with tools for reasoning, shape validation, and federated processing to ensure the integrity and usability of the graph structure. These capabilities are essential for structuring complex biomedical knowledge accurately, providing a foundation for reliable AI-driven insights in clinical applications.



(b) **Knowledge Graph Analytics and Human-Machine Communication in TrustKG.** The Knowledge Graph Analytics layer combines symbolic learning, numerical learning, and causal analysis, enabling tasks such as valid link prediction and counterfactual prediction. The Human-Machine Communication layer provides a natural language interface and integration with scientific publications, making it easier for clinical users to interpret AI-driven recommendations. Thus, enabling TrustKG to offer interpretable and actionable insights, bridging the gap between complex data analytics and real-world clinical decision-making, and demonstrating the practical value of semantic AI in healthcare.

FIGURE 2. TrustKG Framework: This architecture allows TrustKG to deliver accurate, interpretable, and clinically relevant insights, supporting evidence-based decision-making in complex medical settings.

C. PREDICTION PROBLEMS IN TRUSTKG

TrustKG addresses two core prediction tasks critical in oncology: *Valid Link Prediction* and *Counterfactual Prediction*. These tasks are of great importance for the implementation of personalized treatment strategies and the anticipation of patient outcomes in complex scenarios.

1) VALID LINK PREDICTION

This task involves assessing the likelihood that a given KG link is accurate. In lung cancer, this can mean predicting the probability of relapse based on biomarkers and other patient-specific factors. Accurate link prediction enables early intervention and improved prognosis by identifying

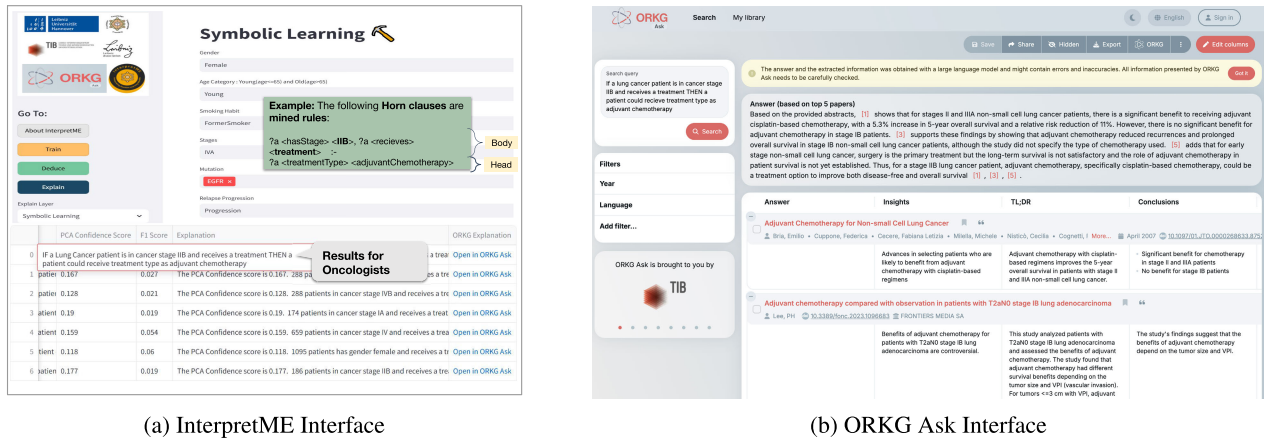


FIGURE 3. Human-Machine Communication in TrustKG. (a) InterpretME Interface: The interface displays mined Horn rules (e.g., treatment recommendations based on cancer stage and mutation type) with the corresponding confidence scores and explanations. These results are transformed into natural language, making them accessible to oncologists for interpretation and decision-making. Additionally, links to ORKG explanations provide further insights into the derived rules. **(b) ORKG Ask Interface:** The ORKG Ask system retrieves relevant scientific publications and presents query results (e.g., chemotherapy treatment outcomes) along with concise summaries, insights, and links to full articles. This enables oncologists to access evidence-based findings directly connected to the mined rules, supporting informed decision-making.

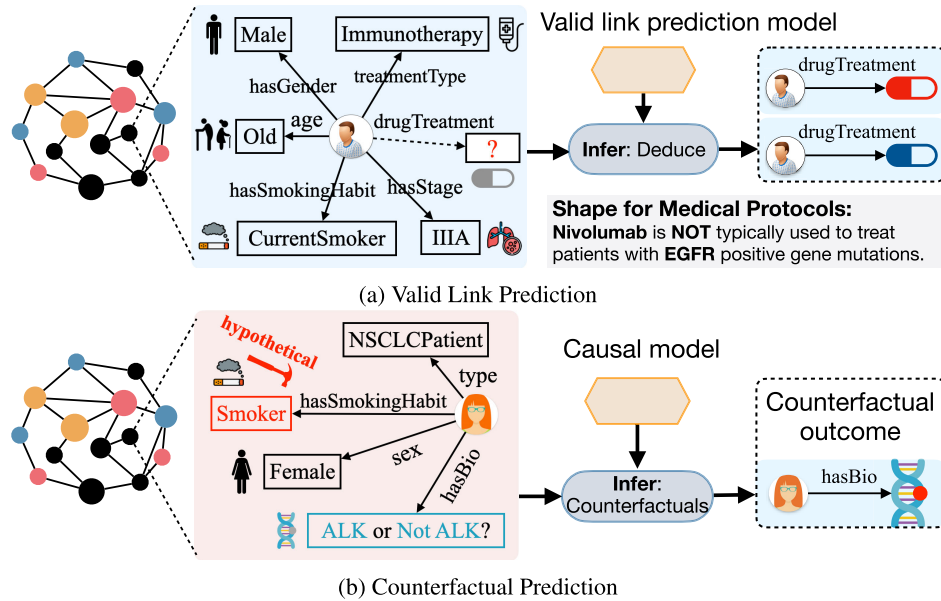


FIGURE 4. Prediction Tasks in TrustKG. (a) Valid Link Prediction: An example illustrating the identification of a link between patient attributes and treatment protocols. The model deduces whether the prescribed treatment aligns with clinical guidelines, indicating if it is valid for a patient with specific characteristics (e.g., age, gender, and genetic mutations). **(b) Counterfactual Prediction:** An example of counterfactual reasoning applied to evaluate how a hypothetical scenario (e.g., change in smoking habits) might impact the patient's biomarker profile, supporting personalized treatment decisions and preventive care.

at-risk patients. In the context of lung cancer, *Valid Link Prediction* is illustrated by assessing whether a prescribed treatment, such as immunotherapy, is valid for a specific patient profile. The example in Figure 4a shows a patient with attributes including age, gender, smoking habits, and cancer stage. The predictive model evaluates if the patient's drug treatment aligns with medical protocols, such as whether Nivolumab is recommended for patients without EGFR-positive mutations. By anticipating the efficacy of this treatment link, TrustKG facilitates the delivery of optimal care, promoting early intervention and enhanced outcomes.

2) COUNTERFACTUAL PREDICTION

Counterfactual reasoning predicts outcomes under hypothetical scenarios, providing a “what-if” analysis. In lung cancer, counterfactual predictions help evaluate how changes in lifestyle factors, like smoking cessation, might influence disease progression. This capability is essential for personalized treatment planning and preventive care. Counterfactual Prediction enables “what-if” analyses by predicting outcomes under hypothetical scenarios. In the example shown in Figure 4b, the model examines how changes in lifestyle factors, such as smoking habits, might influence a biomarker

status related to lung cancer. For instance, it assesses if switching from a current smoker to a non-smoker could affect the ALK mutation status. This analysis supports clinicians in evaluating the potential benefits of lifestyle changes, aiding more in personalized treatment planning, emergency response planning, and preventive care by providing a deeper understanding of patient-specific risks.

TrustKG integrates two specialized hybrid AI systems within: **VISE** for valid link prediction and **HealthCareAI** for counterfactual prediction. Both systems are grounded in a clear specification, leveraging a boxology of design patterns to synergistically combine symbolic reasoning with inductive learning over KGs. **VISE** supports relapse risk assessment by uncovering latent connections, while **HealthCareAI** enables practitioners to explore alternative treatment scenarios tailored to individual needs. Together, **VISE** and **HealthCareAI** illustrate the potential of TrustKG to improve decision-making in healthcare by providing interpretable and personalized insights that support both immediate and preventive medical actions.

IV. PREDICTING VALID LINKS OVER KGs

This section defines the problem of valid link prediction and presents **VISE**. The boxology of design patterns proposed by van Bekkum et al. [54] is utilized to specify the main components of **VISE**. Further, the impact of considering semantics encoded in integrity constraints is empirically evaluated, and the experimental results are reported. The following fundamental concepts will be summarized: *shapes*, *shapes schemas*, *inductive learning*, *Partial Completeness Assumption* heuristic and score, and *Design Patterns*.

Shapes and Shapes Schemas: *Shapes* represent sets of conditions that nodes or edges must satisfy to ensure data consistency and integrity in KGs. A shape, denoted as ϕ , can include basic conditions like truth values, membership tests, Boolean conditions, conjunctions, negations, or cardinality constraints on edges. *Shapes schemas*, denoted as $\Sigma = (\varphi, S, \lambda)$, organize these shapes into a framework consisting of a set of shapes (φ), labels (S) for identification, and a mapping function (λ) linking labels to shapes. Evaluation of a shape determines whether a node or edge satisfies the defined constraints, yielding a binary result (0 or 1). A KG satisfies a shapes schema if all nodes targeted by the schema validate their respective shapes. This structured approach provides a foundation for encoding and enforcing constraints in KGs.

Inductive Learning Over KGs: Inductive learning in Knowledge Graphs (KGs) derives general patterns from specific data to make predictions about unseen facts [51]. It employs methods such as *Knowledge Graph Embeddings*, which map KGs to low-dimensional vector spaces to capture relationships numerically, and *Symbolic Learning*, which identifies logical rules (e.g., Horn clauses) denoted as R for symbolic reasoning. To address the incompleteness nature of KGs, the *Partial Completeness Assumption* (PCA) heuristic integrates observed positive edges (E^+) with *heuristic-based negative edges* (hE^-). These hE^- edges represent plausible

but unobserved relationships entailed from the logical structure of rules. The *PCA Confidence Score*, $PCA(R) = \frac{\text{support}(R)}{|E^+ \cup hE^-|}$ measures a rule's reliability by balancing its support based on edges entailed from R , against the union of observed (E^+) and entailed edges (hE^-).

Boxology of Design Patterns: Van Bekkum et al. [54] introduced modular design patterns, known as “boxology,” to describe hybrid neuro-symbolic systems using elementary design patterns. These patterns visually represent system components, with inputs/outputs as grey rectangles, processes as blue rounded rectangles, models as yellow hexagons, and actors as green triangles. Key patterns for generating models include: *Training*, where data or symbols are used to create a model; *Transformation*, where data or symbols are converted into new forms without creating a model; and *Inference for Symbols*, where a model deduces symbols from data or symbols. These patterns are applied to the proposed solutions for predicting valid links and deducing counterfactual prediction (section V).

A. PROBLEM STATEMENT

Given a directed edge-labeled graph $KG=(V, E, L)$ where each node $v \in V$ represents an entity and each $p \in L$ represents a unique relation between entities, the task of link prediction focuses on completing an incomplete triple $\langle s, p, ? \rangle$. Specifically, this involves identifying the most plausible entity $o' \in V$ that completes the triple (s, p, o) , such that both s and o' validate a given shape schema $\Sigma = (\varphi, S, \lambda)$. Using a scoring function $\theta(s, p, o')$, which quantifies the plausibility of the triple, the optimization problem is defined as:

$$o' = \arg \min_{e \in V} \theta(s, p, e) \wedge s \models \varphi \wedge e \models \varphi$$

The goal is to infer the most plausible entities o' by leveraging positive edges E^+ , generating heuristic-based negative edges hE^- , and ensuring that the resulting triples satisfy the shape schema Σ , i.e., $(s, p, o') \models \varphi$.

B. PROPOSED SOLUTION

The present study puts forth a novel approach, termed **VISE**, for addressing the challenge of *Valid Link Prediction*. **VISE** follows the hybrid design pattern as illustrated in Figure 5, strategically combining numerical learning with symbolic learning and constraints validation methods. Figure 5a illustrates *Symbolic learning* component which is applied to the input KG, resulting in the generation of logical rules and PCA heuristic-based edges. For instance, a logical rule, *relapseProgression*(?a, *Progression*), *drug*(?a, *Nivolumab*) $\Rightarrow$ *biomarker*(?a, *EGFR_Negative*) stating that if a patient has prescribed nivolumab in the progression is more likely to have EGFR Negative mutation. The learned heuristic-based edges serve as prior knowledge, improving numerical learning approaches such as KGE models combined with constraints validation and KG transformation. During the process of symbolic learning, **VISE** utilizes

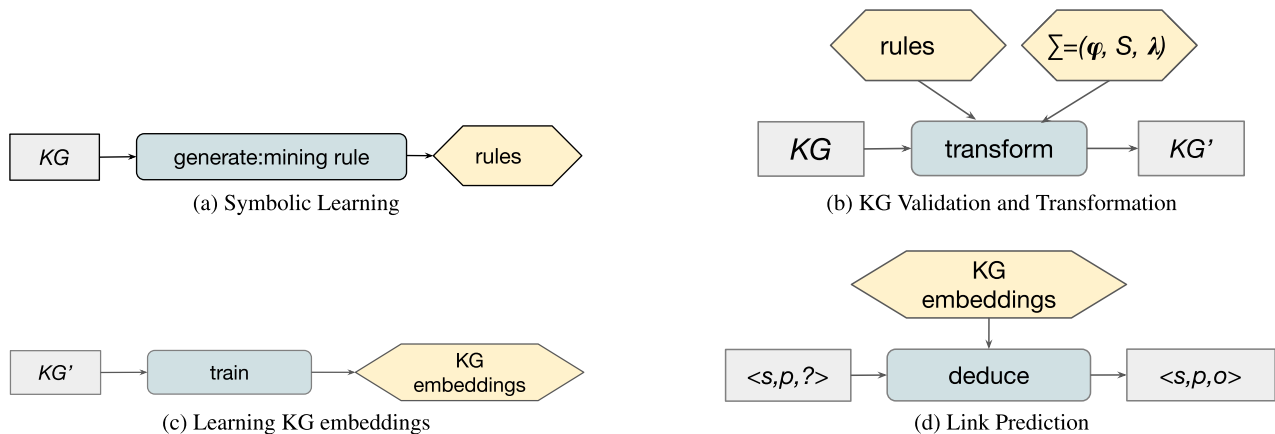


FIGURE 5. **VISE** (corresponds to the Numerical Learning component in Figure 2). Hybrid design pattern (Figure 5a to 5d), demonstrates the use of symbolic rules from the Symbolic Learning component and Constraint Validation and KG Transformation component in combination with Numerical Learning (i.e., KGE models) to enhance the predictive performance.

extracted Horn rules in conjunction with PCA Confidence to deduce heuristic-based negative edges. The mined rules are subsequently utilized to generate predictions regarding the missing relationships in the input KG. These predictions are based on logical inference, which is used to calculate the entailment of the mined rules. SPARQL queries are employed to infer the entailment of mined horn rules and generate heuristic-based negative edges (hE^-) which represent the deduced knowledge based on the observed explicit patterns.

The predictions generated by the symbolic learning system in conjunction with the input KG are then fed to the *KG Validation and Transformation* (Figure 5b) component, where the predicted links are evaluated to determine the validation or invalidation outcomes for SHACL constraints. Here, the validation framework is defined by $\Sigma = (\varphi, S, \lambda)$, a symbolic component to ensure the quality of the predicted hE^- edges. For instance, a SHACL constraint is to check *if a patient mutated with EGFR is not recommended Afatinib drug*. Furthermore, the generated validation report is utilized to transform or rewrite the KG with categorical representation to contain the symbolic knowledge resulting from the constraints validation. The transformed KG is then provided as input to the numerical learning models, i.e., KG embedding models, during the training phase (Figure 5c). This is achieved by processing the data into a low-dimensional vector space. The process of numerical learning is capable of predicting missing links, thereby completing the KGs by deducing links that validate the constraints at a higher rank and with a greater probability of accuracy (Figure 5d). *Categorical representation of KGs* before giving as input to the numerical learning component transforms the KGs to contain negated facts, allowing the KGE model to learn all the representations, enhancing the performance of the models and empowering KG completion. Several studies demonstrated the need for negated facts in KGs to boost the performance of KGE models. **VISE** employs a two-fold rewriting process. Firstly, it evaluates the links predicted by symbolic learning using constraints. Second, depending upon

the validation report of the predicted link. If the patient in the lung cancer KG invalidates the constraint, the links that resemble the patient characteristics in the KG are added with negation, i.e., negated facts. Furthermore, each component builds upon the previous one, creating a fully-fledged hybrid framework, **VISE**, that encompasses enrichment, validation, and transformation of deduced knowledge based on observed explicit patterns to infer meaningful predictions.

Figure 6 illustrates each of the **VISE** components. During *Symbolic Learning* (a), where mining rules are generated. In this case, a rule suggests that patients receiving *Nivolumab* for *stage IIIA* lung cancer as part of *immunotherapy* are more likely to exhibit certain biomarkers, such as *EGFR Negative*. In the *KG Validation and Transformation* step (b), SHACL constraints are applied to validate predicted links. For instance, a constraint checks whether patients with an *EGFR* mutation are recommended against *Nivolumab*. The rules and constraints are used to transform the initial KG into an enriched KG' with both positive and heuristic-based negative edges. The *Learning KG Embeddings* step (c) trains a KG embedding model on the transformed KG', capturing both validated and negated relationships in a low-dimensional representation. Finally, in the *Link Prediction* phase (d), the embeddings are used to deduce new plausible links, such as predicting the likelihood of a patient having *relapse progression* based on existing patterns. This pipeline shows how **VISE** combines symbolic rules, constraints validation, and numerical learning to enhance prediction accuracy in KGs.

C. EXPERIMENTAL STUDY

The effectiveness of **VISE** for the valid link prediction (VLP) task is assessed using a synthetic Lung Cancer (SLC) KG. The VLP task aims to predict relationships, such as determining whether a patient with lung cancer is in relapse (e.g., (PatientID, hasRelapse, ?)). The study addresses three main research questions: **RQ1**) How does KG transformation using symbolic rules and integrity constraints

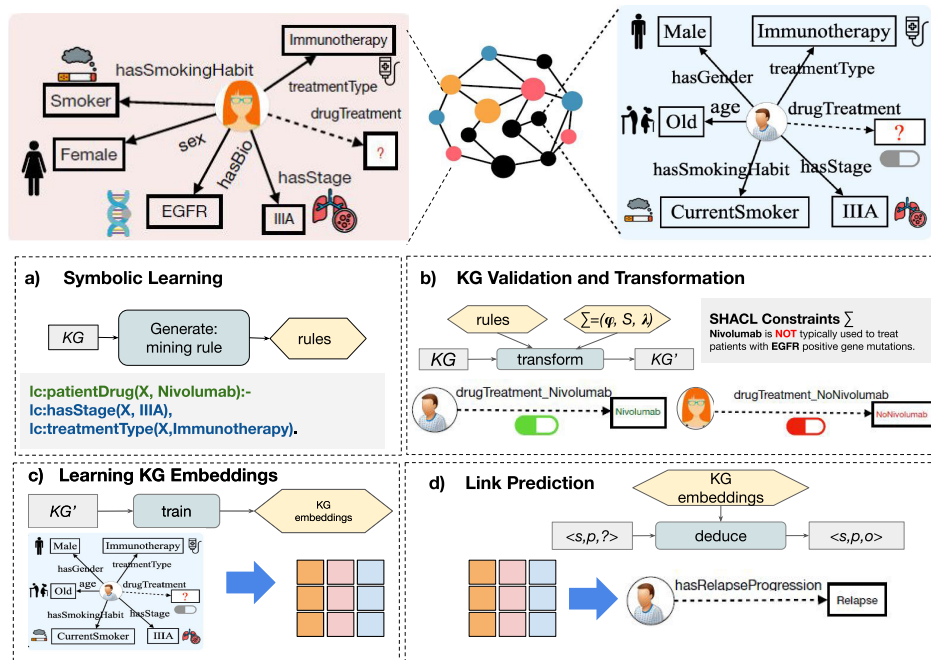


FIGURE 6. Valid Link Prediction Example. Figure demonstrates VISE, starting from symbolic learning to the link prediction step. a) Generates mining rules, shown with an example rule about prescribing Nivolumab for stage IIIA lung cancer patients receiving immunotherapy. b) Applies rules and shapes Σ to transform the initial KG into KG', containing positive and negative facts. c) KG' as an input to learn embedding representation. d) Uses KGE models to predict new relationships.

TABLE 1. Benchmark Statistics represents triple counts along with counts of entities, relations, and existing known-true-and-false facts.

Synthetic Lung Cancer (SLC) KG			
#triples	20581	#known-true	200
#entities	383	#known-false	170
#relations	43		

affect VLP accuracy? **RQ2**) How do KG size and edge variety (true/false) influence VLP performance? **RQ3**) How does VLP compare with classification tasks implemented using AI models learned from relational data?

1) BENCHMARK

The effectiveness of VISE is evaluated using a synthetic Lung Cancer KG (SLC KG) that is derived from the KGs reported by [2]. Table 1 summarizes the statistics of the SLC KG, which contains anonymized characteristics of patients diagnosed with lung cancer. These records include medical history and characteristics such as smoking status (e.g., *non-smoker*), cancer stage (e.g., *stage IV*), demographics, cancer mutation types (e.g., *PD-L1 positive*), proposed treatments (e.g., *afatinib*), relapse status (e.g., *relapse* or *no relapse*), and treatment types (e.g., *chemotherapy*). The prediction problem focuses on determining the *Relapse* status of a lung cancer patient, categorized as either *Relapse* or *No Relapse*. SHACL constraints are used as medical protocols to specify when certain drugs should be prescribed based on a patient's mutations. For example, a protocol might state: "If a patient is mutated with EGFR positive, usage of Nivolumab is not

recommended". The SLC KG is enriched with a schema containing four SHACL constraints.

2) BASELINES

In this study, a systematic approach was employed to evaluate and compare four distinct baselines. **Baseline 1** comprises assessing the state-of-the-art KGE models for the KG completion task. **Baseline 2** shows the impact of transformed KG with KGE models. **Baseline 3** uses the hybrid approach, SPARKLE [55], which improves the performance of KGE models by utilizing symbolic learning approaches. SPARKLE approach is combined with patient outcomes that meet the medical protocols in **Baseline 4**. VISE integrates the fusion of SPARKLE with the transformed KG including validation and violation results to enhance the performance of KGE models. The current VISE implementation employs various state-of-the-art KGE models from the PyKEEN [56] framework, which includes translation and rotational models such as TransE [57], TransD [58], TransH [59], and RotatE [60]. Translation-distance space models, including TransE, TransD, and TransH, translate the head entity's geometric embedding space with a given relation closer to the tail entity. A model for learning embeddings in Euclidean and hyperplane space, RotatE, has received attention for learning symmetric-and-asymmetric properties, and relationship types such as 1:1, 1:N, and M:N.

In addition, an effort is made to compare the predictive capabilities of a traditional machine learning (ML) model—namely, random forests and decision trees [23] with KGE

models in the tasks of predicting the relapse of a lung cancer patient. To assess these models, a hybrid framework, InterpretME [61], is utilized. This framework encompasses the training of machine learning (ML) models and provides human-and-machine understandable interpretability of the ML models' outcomes. In this study, three configurations will be considered: the absence of sampling, undersampling, and oversampling. Without corresponds to the baseline setting where ML models are trained using the original, unmodified target class distribution, i.e., no sampling strategy is applied. In Undersampling, the strategy corresponds to the data is balanced by reducing samples from majority classes to match the number of samples in minority classes. Conversely, oversampling corresponds to achieving balance by increasing the number of samples in minority classes to match the count of majority classes. The traditional ML models are assessed in terms of Precision (P), Recall (R), and F1-score (F1).

3) IMPLEMENTATION

VISE is implemented in a virtual machine on Google Colab with 40 GiB VRAM and 1 GPU NVIDIA A100-SMX4, with CUDA version 12.2 (Driver 535.104.05) using Python 3.10. The VISE implementation code, the benchmark SLC KG, and the trained KGE models are openly accessible in the public GitHub repository.² Figure 5 demonstrates the hybrid design pattern, integrating inductive learning with symbolic learning techniques. Symbolic learning includes logical horn rules (R) and SHACL constraints (ϕ). Symbolic learning is performed over the input KG, resulting in rules, heuristic-based edges, and SHACL validation. Thus, the inferred heuristic edges with validation results are utilized as implicit knowledge to enhance inductive learning, i.e., KGE models. The predictions generated from the symbolic rules and constraints materialized in the input KG and fed as input to inductive learning. The benchmark KGs are divided into 80-20 train-test splits. The model's efficacy in the LP problem is evaluated using metrics such as $Hits@K$ and MRR proposed by Akrami et al. [62]. Both metrics have values between 0 and 1, and higher conveys better. To avoid overfitting, the default settings for the training KGE models include a learning rate of $1e^{-1}$, and *Adam* as the regularization optimizer with a negative sampling strategy. VISE relies on [15] and [55] for symbolic learning methods. Furthermore, the proposed approach is model-agnostic and compatible with other symbolic and inductive learning approaches. InterpretME [30], [31], [33] is employed to execute the traditional ML models such as Decision Trees (DT) and Random Forest (RF). InterpretME as a pipeline offers data integration, curation, and hyper-parameter optimization essential for training the ML models. Moreover, in comparison to KG embeddings, the train-test split ratio is the same for traditional ML models. The predictive pipeline utilizes cross-validation (CV) [63] k -folds stratified shuffle split strategy, i.e., 5-folds. The performance of the predictive models is

TABLE 2. KG Evaluation. Empirical evaluation of KGE models on SLC KG. Hits@1, Hits@3, Hits@5, Hits@10 and MRR are reported. Four baselines and VISE (in light green color) indicate the impact of considering semantics in prediction tasks. The values in bold convey better results and the values in underlined correspond to the best results among the baselines.

Approaches	Model	Results for Synthetic Lung Cancer KG				
		Hits@1	Hits@3	Hits@5	Hits@10	MRR
Baseline 1	TransE	0.000	0.560	0.795	0.943	0.324
	TransD	0.002	0.551	0.690	0.872	0.310
	TransH	0.622	0.864	0.943	0.983	0.756
	RotatE	0.696	0.933	0.969	0.987	0.820
Baseline 2	TransE	0.000	0.713	0.840	0.931	0.376
	TransD	0.008	0.694	0.824	0.935	0.379
	TransH	0.882	0.969	0.997	1.000	0.929
	RotatE	0.864	0.987	0.995	1.000	0.924
Baseline 3	TransE	0.000	0.519	0.747	0.923	0.310
	TransD	0.011	0.551	0.716	0.884	0.322
	TransH	0.596	0.876	0.925	0.977	0.740
	RotatE	0.714	0.941	0.969	0.990	0.829
Baseline 4	TransE	0.000	0.536	0.735	0.931	0.311
	TransD	0.002	0.551	0.733	0.870	0.318
	TransH	0.542	0.849	0.908	0.974	0.702
	RotatE	0.700	0.945	0.972	0.992	0.818
VISE	TransE	0.000	0.760	0.878	0.948	0.388
	TransD	0.013	0.684	0.762	0.884	0.368
	TransH	0.868	0.980	0.994	1.000	0.924
	RotatE	0.887	0.986	0.996	0.998	0.936

evaluated in terms of metrics such as Precision (P), Recall(R), and F1-score (F1). Recall depicts the proportion of counts of correctly predicted patients in the RelapseProgression (RelProg) class to the total patient count with the target class Relapse in the benchmark. Precision is the ratio of accurately predicted patients in the RelapseProgression class to those projected to have class Relapse. The same evaluation parameters are used to categorize lung cancer patients as having NoRelapseProgression (No_RelProg).

The present study investigates the efficacy of VISE, with a particular emphasis on KGE models (TransE, TransD, TransH, and RotatE) in the context of predicting relapses in lung cancer. Table 2 showcases the comparison between baselines and VISE for link prediction. The analysis revealed a robust performance compared to baselines. KGE models are trained over the LC KG, i.e., positive edges E^+ , to predict missing links. The evaluation report presented in Table 2 is obtained using the optimized hyperparameters provided by the PyKEEN pipeline. The impact of positive and negated facts is assessed with Hits@1, Hits@3, Hits@5, Hits@10, and MRR in valid link prediction. TransE, a basic translation model, emerged as performing worst in all baselines with benchmarks respectively. Nevertheless, highlighting the limitations of TransE in modeling 1:N relationships leads to poor performance, particularly in predicting the correct tail at the topmost position. TransH model results support the claim in [59], that it outperforms TransE and TransD models. In SLC KG, TransH performance contributes to promising results in capturing complex geometric relationships with Hits@1 score values ranging from 0.622 to 0.868. TransD, which uses relation-specific projections to translate the embedding space, yields slightly lower values than TransH and TransE. However, RotatE indicates the best performance in all the testbeds except in **Baseline 2**. In **Baseline 1** and **VISE**, the values of Hits@1 range from 0.696 to 0.887. It has been demonstrated that the evaluation of the

²<https://github.com/SDM-TIB/VISE>

benchmark KG in different experimental testbeds, referred to as **VISE**, exhibits superior performance in comparison to other baseline approaches. The experimental evaluation comprises 20 testbeds per baseline, amounting to a total of 100 testbeds. In summary, the results of the evaluation demonstrate the effectiveness of TransH and RotatE for knowledge graph completion in the context of lung cancer relapse prediction.

However, the rationale behind the KGE models may be difficult to understand. The experimental results demonstrate the need for explanations and assistance to understand KGE model behavior. **VISE** shows improved KGE model performance and provides two types of post hoc explanation for the prediction problem. In **VISE**, KGE models showed marginally better performance compared to **Baseline 1**. The heuristic-based negative edges (hE^-) generated by symbolic learning demonstrate the effectiveness of enhancing the performance of **VISE**. The addition of hE^- edges to KG has been deemed a sufficient rationale, as evidenced by the improved performance of the KGE model in terms of Hits@K and MRR. For example, Table 5 displays examples of mined Horn rules that were chosen based on the SHACL constraints, i.e., clinical guidelines used to infer the hE^- edges. Further, the removal of these edges from KG resulted in a notable decline in performance, which can be attributed to the necessity of these facts in the prediction performance thereby answering **RQ1**.

The Horn rules are mined using AMIE [64] over SLC KG; rules are selected based on biomarkers, medications, and therapies, and ranked according to the *PCA confidence* score. The effectiveness of **VISE** is evaluated in terms of the impact of validating constraints for the missing link being predicted by the symbolic learning technique. The heuristic-based negative edges (hE^-) are predicted using the Partial Completeness Assumption (PCA) heuristics from the input KG. The PCA Confidence of a Horn rule, which indicates the amount of incompleteness in a KG, is employed to infer new links and predictions. These predictions are validated by applying the SHACL constraints to determine the validity of the inferred links. The results demonstrated in Table 5 indicate the amount of valid and invalid predictions produced by the symbolic learning techniques. Table 5 shows examples of the symbolic rules, for example, $\text{stage}(\text{?a}, \text{IV}), \text{treatment}(\text{?a}, \text{Immunotherapy}) \Rightarrow \text{drug}(\text{?a}, \text{Nivolumab})$ stating that if a stage IV lung cancer patient received *Immunotherapy* treatment then it is more likely that the patient receives Nivolumab is with the *PCA Confidence* score of 0.83. As mentioned before, the heuristics-based negative edges (hE^-) or predictions are validated using SHACL constraints, and Table 5 shows the number of valid ($\#v$) and invalid ($\#in$) links for each of the SLC KG used as a benchmark in **VISE**. Consequently, the impact of symbolic rules and constraints utilized to explain the KGE models is demonstrated, thereby enabling an answer to be provided to the **RQ2**.

TABLE 3. Decision Tree (DT). The table presents a DT-generated classification report for *RelapseProgression (RelProg)* and *No RelapseProgression (NoRelProg)*, including Precision (P), Recall (R), and F1-score (F1).

DT Classes	Without			Undersampling			Oversampling		
	P	R	F1	P	R	F1	P	R	F1
RelProg	0.83	0.99	0.90	0.66	0.66	0.66	0.64	0.61	0.62
No_RelProg	0.57	0.04	0.09	0.64	0.65	0.65	0.64	0.66	0.65

TABLE 4. Random Forest (RF). The table presents a RF-classification report for *RelapseProgression (RelProg)* and *No RelapseProgression (NoRelProg)*, including Precision (P), Recall (R), and F1-score (F1).

RF Classes	Without			Undersampling			Oversampling		
	P	R	F1	P	R	F1	P	R	F1
RelProg	0.82	1	0.90	0.68	0.71	0.70	0.67	0.71	0.69
No_RelProg	0.00	0.00	0.00	0.70	0.67	0.68	0.70	0.65	0.67

D. TRADITIONAL ML VERSUS KGE MODELS

Experimental evaluation of traditional ML models and KG embeddings reveals explicit complex patterns of performance and predictive capabilities. The objective of this study is to evaluate the research question (**RQ3**). Tables 3 and 4 show the classification outcomes for Decision Trees and Random Forests in three configurations - Without, Undersampling, and Oversampling. In traditional ML, Table 3 shows Decision Trees (DT) performed best with a precision of 0.83 and a promising recall of 0.99 (F1 score: 0.90) for relapse progression prediction under standard settings, while Table 4 shows Random Forests (RF) showed comparable precision (0.82) and perfect recall (1.00) for similar cases. RF also struggled with no-relapse progression predictions, suggesting limitations in handling class imbalance distributions.

In KG, embedding models evaluated through **VISE** and multiple baselines (Table 2) showed different performance patterns, with RotatE emerging as a consistently strong performer, achieving impressive metrics (Hits@1: 0.887, Hits@3: 0.986, Hits@5: 0.996, Hits@10: 0.998, MRR: 0.936 under **VISE**). The overlap analysis of correctly predicted links in Figure 7 reveals an intriguing complementarity between different inductive learning approaches, with RotatE showing the strongest alignment with decision trees (83.6% overlap, 219 shared predictions), suggesting its ability to capture both traditional feature-based patterns and complex relationships. TransE and RotatE showed substantial prediction alignment (72.5%, 190 shared predictions), indicating RotatE's ability to preserve translational relationships while adding rotational modeling power. The lower overlap between TransE and TransD (54.2%, 142 predictions) indicates that these models capture disparate relationships in SLC KG.

VISE enhances performance across most KGE models, particularly benefiting TransH and RotatE, with TransH achieving perfect Hits@10 scores and strong MRR (0.924). These patterns suggest that while individual models excel in specific aspects, offering complementary strengths: traditional ML models provide interpretable predictions for

Overlap of Correct Predictions Across Models

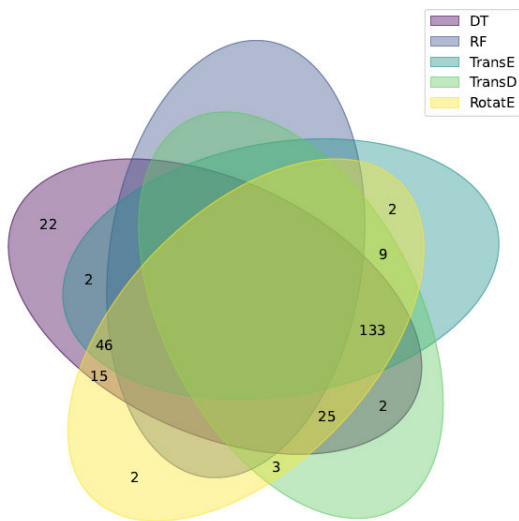


FIGURE 7. Overlap Analysis. Different models for several predictions; the analysis includes: Decision Trees (DT), Random Forest (RF), TransE, TransD, TransH.

straightforward cases, while KGE models, especially RotatE, excel at capturing complex relationships in the KG structure. The varying overlaps despite strong individual performances indicate that different models capture distinct aspects of the underlying relationships, supporting the potential value of ensemble approaches. TransH and RotatE's superior performance under **VISE**, combined with RotatE's strong overlaps with other models, suggests these approaches might serve as robust foundations for predictive models that require both relationship modeling capability and traditional feature-based prediction power.

E. DISCUSSION AND LESSONS LEARNED

The observed results indicate that existing KG embedding (KGE) models are significantly influenced by how factual statements are represented in KGs, with performance improvements observed when valid and invalid links are explicitly defined according to domain-specific integrity constraints. This evidence supports that the hybrid approach implemented in **VISE** effectively leverages the semantics of integrity constraints and mined rules to improve the predictive accuracy of KGE models, providing a more precise solution to the valid link prediction problem. Additionally, as described in subsection III-A, TrustKG facilitates human-machine communication by translating mined rules into natural language and retrieving relevant publications from ORKG ASK. Table 5 presents examples of mined rules, natural language explanations, and related scientific papers. These results also highlight the role of the hybrid methods in **VISE** in improving the interpretability of its results.

V. COUNTERFACTUAL PREDICTION

This section defines the problem of counterfactual and presents **HealthCareAI**. The boxology of design patterns

proposed by Van Bekkum et al. [54] is also used to specify the main components of **HealthCareAI**. The following fundamental concepts will be introduced: *causal analysis*, *causal relationships*, and *causal models*.

Causal Analysis: Causal analysis in KGs examines the effects of interventions or treatments on outcomes. An entity in the KG, known as a *unit*, undergoes an intervention represented by $do(T = t)$, where T is the treatment variable and t is a specific treatment value. It distinguishes between observed (factual) outcomes and hypothetical (counterfactual) outcomes under alternative treatment scenarios, enabling reasoning about cause-and-effect relationships in KG.

Causal Relationships: Causal relationships in KGs are represented using several models: i) *Causal Graphs (CGs)*: Directed acyclic graphs (DAGs) that depict direct causal relationships between variables. In a causal graph $G = (X, E^c)$, a directed edge $(X_i, X_j) \in E^c$ indicates that X_i has a causal influence on X_j . ii) *Causal Bayesian Networks (CBNs)*: Extending causal graphs, CBNs encode probabilistic relationships and intervention effects among variables. The joint probability distribution $P(X)$ over X can be factored based on the graph structure. When an intervention $do(T = t)$ occurs, the interventional distribution $P_t(X)$ computed adjusting the influence of other variables on T .

Causal Models: Causal models in KGs map treatments and contexts to outcomes, predicting how interventions affect specific entities. For a KG $KG = (V, E, L)$, key concepts include: i) A *target class* C , representing the type of entities (e.g., patients) being analyzed. ii) A *treatment property* p_T and an *outcome property* p_Y associated with class C . iii) *Contextual properties* P_C^{ctx} , representing additional factors related to C that may influence the outcome.

For an entity e of class C , the treatment is defined by (e, p_T, t) and the outcome by (e, p_Y, y) . The context consists of all the triples involving properties in P_C^{ctx} . A causal model ϑ maps a treatment and its context to an outcome as $\vartheta((s, p_T, t), E_{ctx}) = (s, p_Y, y)$, where E_{ctx} contains the contextual information for s . The unit dataset $D_{KG}(C)$ aggregates entities, treatments, outcomes, and contexts of type C , enabling causal effect estimation across similar units.

A. PROBLEM STATEMENT

Given a Knowledge Graph $KG = (V, L, E)$, a target class C with its properties P_C , including treatment p_T , outcome p_Y , and contextual properties P_C^{ctx} , the goal of counterfactual prediction is to determine the outcome for hypothetical treatments. Let $V_t = \{o \mid (s, p_T, o) \in E\}$ represent possible treatment values. For a unit e of type C , consider a hypothetical treatment (e, p_T, t') where $t' \in V_t$ but $(e, p_T, t') \notin E$. The objective is to find an optimal *causal model* ϑ^* for predicting the outcome p_Y under this hypothetical treatment. This optimal model ϑ^* is selected from the space of all possible causal models Θ to maximize a utility function $f(., .)$

TABLE 5. Exemplary mined horn rules.

Exemplary Mined Horn Rules	Natural Language Statements	ORKG ASK	PCA Conf.	hE #v	#in
drug(?a, Nivolumab) $\Leftarrow$ stage(?a, IV) treatment(?a, Immunotherapy)	IF a Lung Cancer patient is in cancer stage IV and receives treatment type as immunotherapy THEN a patient could receive Nivolumab drug	Ask 1	0.83	50	40
biomarker(?a, EGFR Negative) $\Leftarrow$ relapseProgression(?a, Progression) drug(?a, Pembrolizumab)	IF a Lung Cancer patient is in progression and receives Pembrolizumab drug THEN a patient has EGFR Negative mutation	Ask 2	0.97	0	10
drug(?a, Nivolumab) $\Leftarrow$ treatment(?a, Immunotherapy), treatment(?a, Intravenous Chemotherapy)	IF a Lung Cancer patient receives treatment type such as immunotherapy and Intravenous Chemotherapy THEN a patient could receive Nivolumab drug	Ask 3	0.72	130	100
biomarker(?a, EGFR Negative) $\Leftarrow$ biomarker(?a, ALK Negative), treatment(?a, Radiotherapy to Bone)	IF a Lung Cancer patient has ALK Negative and receives treatment type as Radiotherapy to Bone THEN a patient has EGFR Negative mutation	Ask 4	0.92	20	20

that measures the accuracy of the counterfactual prediction:

$$\vartheta^* = \operatorname{argmax}_{\vartheta \in \Theta} f((e, p_T, t'), \phi(e, P_C^{ctx}))$$

$f(., .)$ evaluates how accurately a causal model ϑ predicts the counterfactual outcome based on the treatment and context.

B. PROPOSED SOLUTION

HealthCareAI solves the problem of counterfactual prediction by performing three main tasks: a) Symbolic reasoning; b) Causal model learning; c) Counterfactual prediction (shown in Figure 8). *Symbolic Reasoning* The input (Fig. 8-a) includes a knowledge graph $KG = (V, E, L)$, a set of Horn rules R over the properties in L , a target class $C \in V$, and treatment and outcome properties p_T and p_Y in L . The set R consists of logical rules, such as those defined in RDFS [65] or OWL [66]. The transformation of KG is guided by a semantic model and an axiomatic system that states the meaning of symbols in KG . Query processing and symbolic learning are performed over KG , applying the logical rules in R to generate an enriched KG' that includes deduced facts and domain knowledge [42]. Domain knowledge includes: i) Metadata for the target class C and its properties $P_C \subseteq L$ (e.g., `rdfs:label`, `rdfs:domain`) [67]. ii) A set of additional Horn rules R' , mined through symbolic learning, which meets certain criteria, such as a minimum confidence and PCA confidence score [55]. These mined Horn rules, which may imply causal relationships, form part of the domain knowledge derived from KG' . This knowledge is used to construct a prompt for a large language model (LLM) [68] to query causal relationships between the properties in P_C . Additionally, KG' is used to create a unit dataset $D_{KG'}(C)$, which supports further causal model learning.

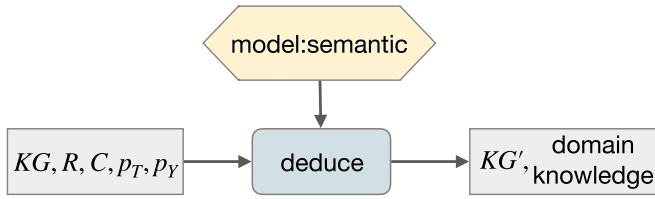
Causal Model Learning (Fig. 8-b) component infers a CG (for the causal model ϑ) over the properties P_C using a hybrid approach. It takes as input the unit dataset $D_{KG'}(C)$ and the LLM prompt from the previous component, and generates a casual model ϑ for counterfactual prediction. To do this, it combines a data-driven statistical model and a knowledge-driven model LLM to infer CG. The statistical model learns a CG $G_1 = (P_C, E_1)$ directly from the dataset $D_{KG'}(C)$. The statistical model can be any traditional causal

discovery method, such as **PC** [69], **FCI** [69], or **GES** [43]. The metadata-driven model (any LLM) takes as input an LLM prompt and returns a CG $G_2 = (P_C, E_2)$. The LLM prompt is designed into four sections, where the role section uses the metadata to provide the domain information of KG and specify the functions of the LLM. The context section uses the metadata of the target class C and properties P_C , including the domain (`rdfs:domain`) and range (`rdfs:range`) of the properties, the human-readable label (`rdfs:label`) and annotation (`rdfs:comment`). In addition, each Horn rule in R' is translated into a set of association pairs (p_1, p_2) , where p_1 and p_2 are the predicates in the head and body of a Horn rule, respectively, as potential causal relationship candidates; the objective section specifies the task of identifying causal relationships between properties in P_C ; the instruction section formats the output of causal relations. The final output of this component is a CG $G = (P_C, E^c)$ s.t. $E^c = E_2 \cup \{(c, e) \in E_1 | (P_C, E_2 \cup \{(c, e)\}) \text{ is a DAG}\}$. In other words, it includes all causal relations in E_2 (by LLM) and those in E_1 that do not introduce any directed circle in G . The causal model ϑ is trained based on G and $D_{KG'}(C)$.

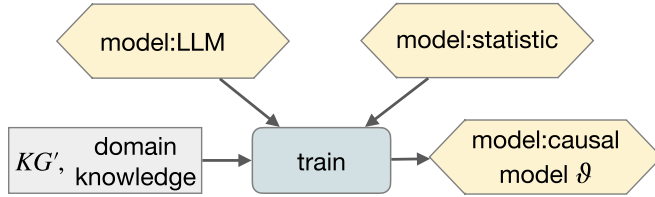
Counterfactual Prediction (see Fig. 8-c) predicts counterfactuals from units. Given a causal model ϑ , a hypothetical treatment (e, p_T, t') and context $\phi(e, P_C^{ctx})$ of a unit e , it predicts the counterfactual on property p_Y of e . In order to establish a proof of concept, the use of CBN as the *causal model* ϑ is employed. Let $\vartheta : (G, P_t)$ be the CBN based on the CG G trained on the dataset $D_{KG'}(C)$. The counterfactual p_Y of e under a hypothetical treatment $(e, p_T, t') \notin E$ is predicted as (e, p_Y, y') s.t. $P_{t'}(y'|x) \geq \max(\{P_{t'}(y|x)\}_{y \in V_Y})$, where $V_Y = \{y | (\exists e' \in V)[(e', p_Y, y) \in E]\}$ contains all unique values of p_Y , and $P_t(y|x) = \frac{\sum_s P_t(y, c, s)}{\sum_{y, s} P_t(y, s)}$ is an interventional distribution (see the definition of CBNs) derived from the CBN model ϑ .

To illustrate, consider a lung cancer patient, *Eva* (depicted on the right side of Fig. 8-a), a non-smoking female with a positive ALK biomarker result. The objective is to predict her biomarker result under the counterfactual scenario where she is a smoker (shown in Fig. 8-c). As a result of symbolic reasoning, **HealthCareAI** extracts domain knowledge from KG , including two parts: (1) metadata stored within KG , and (2) Horn rules mined from an enriched KG' that incorporates

a) Symbolic Reasoning



b) Causal Model Learning



c) Counterfactual Prediction

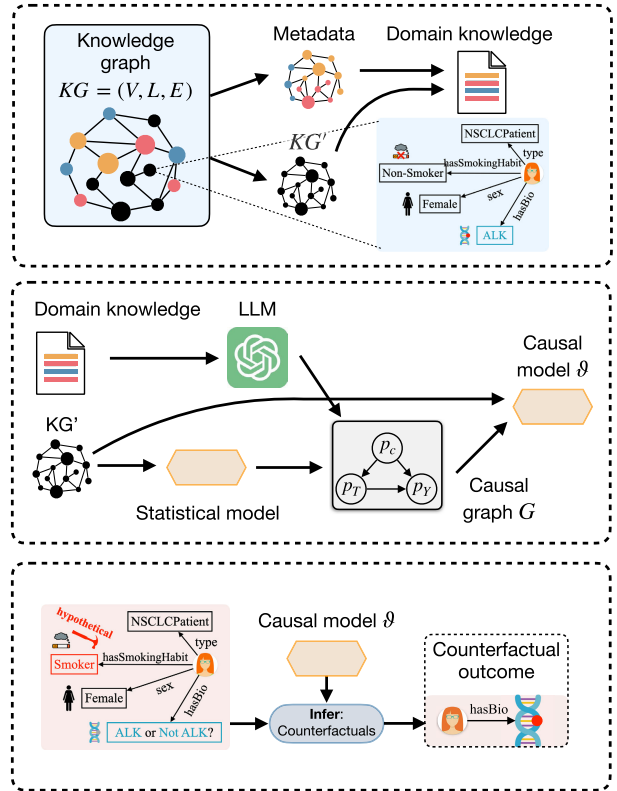
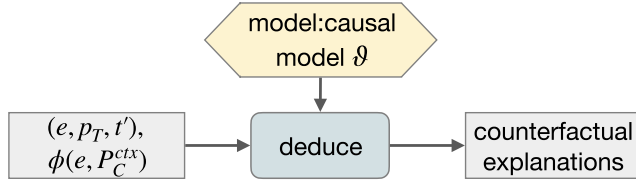


FIGURE 8. HealthCareAI framework (a) **Symbolic Reasoning:** A semantic model deduces an enriched knowledge graph KG' and domain knowledge (metadata and Horn rules) from the input KG , Horn rules R , target class C , and properties p_T and p_Y . (b) **Causal Model Learning:** A causal model ϑ is trained by integrating a statistical model (inferring causal graph G_1 from $D_{KG'}(C)$) and an LLM (producing causal graph G_2 from prompts). The combined graph G ensures acyclicity and supports causal inference. (c) **Counterfactual Prediction:** Using ϑ , hypothetical treatments (e, p_T, t') , and contextual information $\phi(e, P_C^{ctx})$, counterfactual outcomes p_Y are predicted, providing interpretable explanations. The framework integrates symbolic and statistical learning for accurate and interpretable counterfactual reasoning.

implicit facts inferred from KG using a formal system. During the causal model learning phase (see Fig. 8-b), an LLM utilizes the domain knowledge to deduce causal relationships between the properties of the target class—in this case, the NSCLCPatient. In addition, a statistical model directly learns causal relationships from KG' . The inferred relationships are integrated into a single causal graph G using the method specified in the causal model learning component. Subsequently, the causal model ϑ is learned using the data from KG' and G . Finally, the causal model ϑ receives the contextual properties of *Eva* and a hypothetical treatment (i.e., $(Eva, \text{hasSmokingHabit}, \text{Smoker})$) and infers the counterfactual biomarker result of *Eva* if she were a smoker.

C. EXPERIMENTAL STUDY

The efficacy of **HealthCareAI** in causal discovery and counterfactual prediction tasks is evaluated over a synthetic LC KG with varying sizes. These benchmarks are generated from the LC KG reported by Calvo et al. [2]. The study is guided by the following research questions:

RQ4 How can a neuro-symbolic system enhance the discovery of causal relationships in alignment with an expert-defined causal graph?

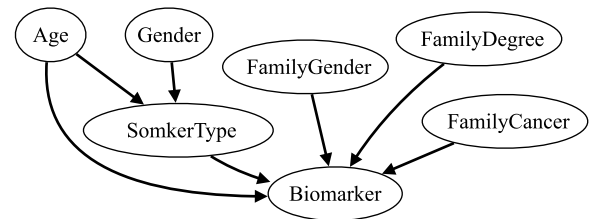


FIGURE 9. Expert Designed Causal Graph G^* built over the properties in synthetic Lung Cancer KG.

RQ5 How does a neuro-symbolic system impact the accuracy and effectiveness of counterfactual predictions?

1) BENCHMARKS

The synthetic KGs include the following properties: Biomarker that is either *ALK* or *EGFR* and other biomarkers, Age that is categorized as *Young* (≤ 50 years) or *Old* (> 50 years), Gender that is *Male* or *Female*, SmokerType that is *Non-Smoker* or *Smoker*, FamilyCancer which is *OnlyMajor* if a patient's family antecedents have only these cancers: *Breast*, *Lung*, *Colorectal*, *Head and neck*, *Uterus/cervical*, *Esophagogastric*, *Prostate*, otherwise *hasMinor*, FamilyGender that is *Women* if all family antecedents are women, *Men* if all of them are men, otherwise

TABLE 6. Benchmarks of synthetic KGs for counterfactual prediction. #CF represents the number of ground truth counterfactuals (triples).

KG	KG _{2k}	KG _{5k}	KG _{10k}
#triples	16180	40180	80180
#entities	2055	5055	10055
#relations	17	17	17
#CF	2000	5000	10000

WomenorMen, and *FamilyDegree* that indicates the degree of relationship of family antecedents to the patient (i.e., first, second, or third degree). In order to train the additive noise model (ANM), it is necessary to utilize the expert-designed CG G^* (see Figure 9) and the LC KG [2]. The ANM is a non-linear model that is used to analyze and predict the behavior of nonlinear systems. The utilization of a trained ANM model, presumed capable of apprehending the underlying causal mechanism inherent within the dataset, facilitates the generation of synthetic KGs comprising varying patient counts, designated as N and ranging from $2k$ to $10k$. Specifically, *Age*, *Gender*, *FamilyGender*, *FamilyCancer* and *FamilyDegree* are simulated from uniform distributions; while *SmokerType*, and *Biomarker* are simulated using logistic functions:

$$Y = 1/(1 + \exp(-(\alpha + \beta \cdot X' + N(0, \sigma^2)))) \quad (1)$$

where Y is the synthetic variable; $X' = Pa_{G^*}(Y)$ include the parent nodes of Y in the CG G^* (see Figure 9). The α and β are learned from the LC KG using a logistic regression. A noise term $N(0, 0.1^2)$ is applied to simulate other potential unobserved factors. Additionally, each synthetic KG includes the metadata of the LC KG. The performance of counterfactual prediction must be evaluated by generating the ground truth counterfactuals on *Biomarker* under intervention on *SmokerType* for each patient using the trained ANM. Given a synthetic KG $= (V, E, L)$, a patient $e \in V$ whose treatment is $(e, p_T, t) \in E$. Let $Y = 1$ denote that *Biomarker* (p_Y) is *ALK* or *EGFR*, otherwise ($Y = 0$) is *other biomarkers*, and $T=t$ denotes the treatments on *SmokerType* (p_T) which are *Non-Smoker* ($t=1$) and *Smoker* ($t=0$). The counterfactual *Biomarker* of a patient (unit) e is generated using the Function 1 with input of $X' = x'$ following the assignment $X'_1, \dots, X'_k := x'_1, \dots, x'_k$ s.t. $(\forall i \in [1, k])[(e, X'_i, x'_i) \in S]$ where $k = |X'|$ is the variable number of X' and S is a set of triples $\phi(e, P_C^{ctx}) \cup \{(e, p_T, t')\}$ s.t. $(e, p_T, t') \notin E$. Table 6 presents the statistics of the synthetic LC KGs, denoted as KG_{2k} , KG_{5k} , and KG_{10k} .

2) BASELINES

HealthCareAI is compared with other baselines, including: **Baseline1**: Statistical models such as **PC**, **FIC**, and **GES** used independently for causal discovery. **Baseline2**: Large language models (LLMs) queried for causal relationships without incorporating domain knowledge (**wo DK**). **Baseline3**: LLMs queried for causal relationships with the inclusion of domain knowledge (**w DK**). **Baseline4**: The

expert-designed causal graph G^* (in Fig. 9) used directly to learn the causal model ϑ .

3) METRICS

In the context of the causal discovery task, the performance of an inferred causal graph (CG) is evaluated against a CG designed by experts (G^*) (in Fig. 9), using the Jaccard Index [70], Precision, Recall, and F1-score [71] to measure the proportion of shared causal relationships between G and G^* . For the purpose of assessing the performance of a system in counterfactual prediction tasks, a comparison is made between the predicted counterfactual results and the true ground truth. Given a list of predicted counterfactuals $\hat{Y}^{CF}$ by a CBN and the ground truth counterfactuals Y^{CF} , the *Pearson correlation coefficient* [72] (PCC) is calculated: $PCC(\hat{Y}^{CF}, Y^{CF}) = \frac{cov(\hat{Y}^{CF}, Y^{CF})}{\sigma_{\hat{Y}^{CF}} \sigma_{Y^{CF}}}$, where $cov(\cdot)$ and σ represent covariance and standard deviation. Higher values indicate better performance.

4) IMPLEMENTATION

The set of logical rules, designated R , is hereby implemented (see Fig. 8-a) as an empty set, since each patient in a synthetic KG has completed properties. The Horn rules R' , as a part of the domain knowledge, are mined from the synthetic KGs using the AMIE [64]. The LLM is implemented by the GPT-4 model [73]. Employing CBNs as the causal model ϑ (see Fig. 8-b) for all methods. CBNs of all methods are implemented using `pgmpy` Python package [74]. The reproduction code and synthetic datasets are available here,³ where the LLM prompts used by **Baseline2**, **Baseline3**, and **HealthCareAI** are presented.

D. IMPACT ON CAUSAL DISCOVERY

The results pertaining to the causal discovery task are presented in Table 7 (for synthetic Lung Cancer KG with $N = 2k$, denoted as KG_{2k}), Table 8 (for $N = 5k$, denoted as KG_{5k}), and Table 9 (for $N = 10k$, KG_{5k}). In general, all methods have better precision performance but relatively weaker recall. This indicates the correct identification of some causal relationships in CG designed by experts G^* (see Figure 9), but the inferred causal edges are not complete. Among, all methods, the **HealthCareAI**, which combines **Baseline3** and **PC** method, outperforms other methods in all synthetic KGs with different settings of N . The data-driven methods, i.e., methods of **Baseline1**, have similar performance compared with the metadata-driven methods (**Baseline2** and **Baseline3**) in small KG (with $N = 2k$); while performance is improved in larger KGs with $N \in \{5k, 10k\}$. The **PC** and **FCI** have exact the same performance across all KGs. This may be explained that both are based on the conditional independence test and there are no hidden confounders in the simulation where **FCI** is more suitable. Even the metrics of **PC** (or **FCI**) are the same in KG_{5k}

³<https://github.com/SDM-TIB/HealthCareAI>

TABLE 7. Comparison of Causal Discovery Performance of Different Approaches Against the Expert Causal Graph. *w* and *wo* denote respectively “with” and “without”, and DK denotes “domain knowledge”.

Approaches	Model	Results (%) for KG_{2k}			
		Jaccard Index	Precision	Recall	F1-Score
Baseline 1	PC	62.5	100.0	62.5	76.9
	FCI	62.5	100.0	62.5	76.9
	GES	37.5	100.0	37.5	54.5
Baseline 2	GPT4 wo DK	40.0	66.7	50.0	57.1
Baseline 3	GPT4 w DK	62.5	100.0	62.5	76.9
HealthCareAI	Baseline 2 + PC	60.0	75.0	75.0	75.0
	Baseline 3 + PC	87.5	100.0	87.5	93.3

TABLE 8. Comparison of Causal Discovery Performance of Different Approaches Against the Expert Causal Graph. *w* and *wo* denote respectively “with” and “without”, and DK denotes “domain knowledge”.

Approaches	Model	Results (%) for KG_{5k}			
		Jaccard Index	Precision	Recall	F1-Score
Baseline 1	PC	75.0	100.0	75.0	85.7
	FCI	75.0	100.0	75.0	85.7
	GES	62.5	100.0	62.5	76.9
Baseline 2	GPT4 wo DK	40.0	66.7	50.0	57.1
Baseline 3	GPT4 w DK	62.5	100.0	62.5	76.9
HealthCareAI	Baseline 2 + PC	70.0	77.8	87.5	82.4
	Baseline 3 + PC	87.5	100.0	87.5	93.3

TABLE 9. Comparison of Causal Discovery Performance of Different Approaches Against the Expert Causal Graph. *w* and *wo* denote respectively “with” and “without”, and DK denotes “domain knowledge”.

Approaches	Model	Results (%) for KG_{10k}			
		Jaccard Index	Precision	Recall	F1-Score
Baseline 1	PC	75.0	100.0	75.0	85.7
	FCI	75.0	100.0	75.0	85.7
	GES	62.5	100.0	62.5	76.9
Baseline 2	GPT4 wo DK	40.0	66.7	50.0	57.1
Baseline 3	GPT4 w DK	62.5	100.0	62.5	76.9
HealthCareAI	Baseline 2 + PC	70.0	77.8	87.5	82.4
	Baseline 3 + PC	100.0	100.0	100.0	100.0

and KG_{10k} , while the inferred CGs in both situations are not exactly the same. Among all methods in **Baseline1**, the **GES** performs the worst. For metadata-driven methods, i.e., **Baseline2** and **Baseline3**, the performance is stable in all settings of N , because these methods do not depend on data. The results of **Baseline3** is constantly better than **Baseline2** which demonstrates the usefulness of domain knowledge (**DK**) in causal discovery. The performance of data-driven methods (**Baseline1**) improves as the data size N increases, but peaks at $5k$ and $10k$. In addition, learning the CG from (observational) data alone may be theoretically impossible [37]. As a complement, the metadata- or knowledge-driven methods (such as **Baseline3**) can infer some causal relationships that cannot revealed by **Baseline1** methods, this explains why **HealthCareAI** can achieve the best performance in all synthetic KGs. These results answer the research question **RQ4** that the neurosymbolic systems enriched with semantics can improve the performance of traditional data-driven methods by using knowledge deduced from KGs.

E. IMPACT ON COUNTERFACTUAL PREDICTION

Based on the CGs estimated from the previous step (subsection V), the CBNs (causal model ϑ) are learned from the dataset $D_{KG}(C)$ over the synthetic LC KGs using the *Maximum Likelihood Estimation* Method [75]. The evaluation results of different methods on counterfactual prediction in various synthetic LC KGs with $N \in \{2k, 5k, 10k\}$ are

TABLE 10. Comparison of Counterfactual Prediction using Causal Models based on Different Causal Graphs produced by different Approaches at $N = 2k$. **Baseline4** use the expert-designed CG G^* . *w* and *wo* denote respectively “with” and “without”, and DK denotes “domain knowledge”.

Approaches	Model	Results (%) for KG_{2k}
		Pearson Correlation
Baseline 1	PC	95.0 (± 2.5)
	FCI	95.0 (± 2.5)
	GES	89.9 (± 1.6)
Baseline 2	GPT4 wo DK	89.9 (± 1.6)
Baseline 3	GPT4 w DK	87.0 (± 1.8)
Baseline 4	Expert causal graph	81.4 (± 2.9)
HealthCareAI	Baseline 2 + PC	93.9 (± 2.7)
	Baseline 3 + PC	81.4 (± 2.9)

TABLE 11. Comparison of Counterfactual Prediction using Causal Models based on Different Causal Graphs produced by different Approaches at $N = 5k$. **Baseline4** use the expert-designed CG G^* . *w* and *wo* denote respectively “with” and “without”, and DK denotes “domain knowledge”.

Approaches	Model	Results (%) for KG_{5k}
		Pearson Correlation
Baseline 1	PC	95.9 (± 0.8)
	FCI	95.9 (± 0.8)
	GES	95.9 (± 0.8)
Baseline 2	GPT4 wo DK	90.7 (± 1.6)
Baseline 3	GPT4 w DK	90.1 (± 1.3)
Baseline 4	Expert causal graph	96.4 (± 0.7)
HealthCareAI	Baseline 2 + PC	96.4 (± 0.7)
	Baseline 3 + PC	96.4 (± 0.7)

reported, respectively, in Table 10, Table 11, and Table 12. The PCC metrics for the CBN by each method are presented as the mean ($\pm$ standard deviation), using 5-fold cross-validation on all patients and corresponding counterfactuals. The results indicate that **PC** and **FCI** of **Baseline1**

TABLE 12. Comparison of Counterfactual Prediction using Causal Models based on Different Causal Graphs produced by different Approaches at $N = 10k$. **Baseline4** use the expert-designed CG G^* . **w** and **wo** denote respectively “with” and “without”, and **DK** denotes “domain knowledge”.

Approaches	Results (%) for KG_{10k}	
	Model	Pearson Correlation
Baseline 1	PC	95.1 (± 0.8)
	FCI	95.1 (± 0.8)
	GES	95.1 (± 0.8)
Baseline 2	GPT4 wo DK	90.0 (± 0.9)
Baseline 3	GPT4 w DK	89.5 (± 0.6)
Baseline 4	Expert causal graph	95.9 (± 0.6)
HealthCareAI	Baseline 2 + PC	95.1 (± 0.8)
	Baseline 3 + PC	95.9 (± 0.6)

outperform others in scenarios with smaller datasets (i.e., KG_{2k}). In contrast, the CBN trained based on the **Expert** CG G^* exhibits the lowest performance, which may be explained by the overfitting issue [76]. This is likely due to the complex structure of the **Expert** CG, which requires a large dataset to learn the conditional probability tables of the CBN. Conversely, the simpler structures of the CGs deduced by the **PC** and **FCI** methods allow for effective learning with small datasets. In the scenario of large datasets (i.e., $N \geq 5k$), the Expert CBN outperforms all CBNs trained based on CGs by other methods. Although enlarging the dataset generally enhances the generalization capability of CBNs, it is notable that the performance of all CBNs slightly declines as N increases from $5k$ to $10k$. The CBNs by the metadata-driven methods, i.e., the **Baseline2** and **Baseline3** perform the worst across all synthetic KGs. This result underscores the limitations of CGs estimated without considering the underlying data, emphasizing the crucial role of data-driven causal graph estimation for robust counterfactual reasoning. By incorporating the causal relationships estimated by the data-driven methods, e.g., **PC**, the CBN produced by **HealthCareAI** achieves competitive performance across all settings of N , compared to the CBN based on the **Expert** CG. The results address the research question **RQ5**, confirming that **HealthCareAI** outperforms the other baselines and exhibits a good counterfactual predicting performance with respect to **Baseline4**, i.e., the one based on the Expert CBN.

F. DISCUSSION AND LESSONS LEARNED

These results show the benefits of integrating semantics and domain-specific knowledge into the causal learning pipelines. By leveraging metadata and mined rules from KGs, **HealthCareAI** improves both the accuracy and interpretability of causal models. Semantic information enables a more structured representation of causal relationships, which not only enhances predictive capabilities but also aligns more closely with human reasoning processes. This hybrid approach also underscores the importance of using enriched domain knowledge to learn accurate causal graphs, supporting detailed counterfactual predictions. Finally, these findings suggest that the explicit representation of semantics within causal models contributes to better decision-making and will facilitate human-machine communication by making results more interpretable and contextually grounded.

VI. LIMITATIONS AND OPEN CHALLENGES IN TRUSTKG

Despite its contributions to enhancing interpretability and reliability in AI systems, TrustKG faces several limitations that must be addressed to fully realize its potential. These limitations stem from inherent challenges in KG-based formalisms and their application frameworks. Key areas requiring improvement include [77]:

- **Fragmentation and Incompleteness.** KGs may comprise sparse data, which hampers their ability to support robust predictive modeling and reasoning. This challenge is particularly evident in data-scarce scenarios, where fragmentation weakens knowledge representation and limits the insights that can be derived.
- **Evolution.** The evolving nature of knowledge across various domains poses significant challenges for KGs in maintaining accuracy and consistency. Frequent updates, whether from emerging research, revised standards, or changing domain-specific guidelines, necessitate adaptive mechanisms to integrate evolving data and knowledge while safeguarding integrity.
- **Usability and Accessibility.** Existing KG tools lack intuitive, user-friendly interfaces and clear visualizations, making them less accessible to users without technical expertise [27]. This issue hinders effective exploration and utilization, reducing their practical impact on decision-making processes across various domains.
- **Transparency and Traceability.** KG-based frameworks often lack robust mechanisms for tracing data origins, documenting reasoning processes, and detailing prediction pipelines [78]. This lack of provenance tracking weakens transparency and accountability, reducing user trust in the generated and recommended insights.
- **Standardized Benchmarking.** The evaluation of KG-based frameworks is limited by the absence of standard evaluation benchmarks to measure the usability, effectiveness, and efficiency of KGs in real-world applications. Without clear methodologies, it is challenging to assess their contributions to decision-making and justify their integration into existing workflows.

Addressing these issues will require advancements in KG development, integration, and evaluation. Developing methods for data enrichment, enabling seamless updates, creating user-centric tools, ensuring traceability, and establishing robust evaluation frameworks are crucial steps toward unlocking the full potential of KG-based systems like TrustKG.

VII. CONCLUSION AND FUTURE WORK

This work has demonstrated the potential of integrating semantics within hybrid AI systems to enhance predictive capabilities, contextual understanding, and interpretability in medical applications, particularly for lung cancer. The proposed framework, TrustKG, along with the hybrid AI

systems **VISE** and **HealthCareAI**, leverages KGs and symbolic reasoning to address critical tasks such as valid link prediction and counterfactual prediction. The results show that semantic integration improves predictive accuracy, as evidenced by the enhanced Hits@1 and MRR scores achieved by **VISE**, and supports more accurate causal discovery, with **HealthCareAI** outperforming baselines on metrics like the Jaccard Index and F1-Score. Additionally, the use of counterfactual reasoning in **HealthCareAI** achieves Pearson Correlation scores close to expert-derived benchmarks, illustrating the effectiveness of integrating domain-specific knowledge. These findings highlight the role of semantic knowledge in advancing AI systems for healthcare by providing more transparent solutions that align with clinical needs. Future research directions include:

Evolving Data Management: One key area is enhancing semantic data management to model evolving and non-monotonic knowledge. This includes handling changes in medical guidelines, emerging treatments, and patient-specific factors over time. Improved models for representing these dynamic knowledge can support AI systems that remain adaptable in clinical practice.

Reducing Computational Costs: While effective, our approach increases execution costs. Future work should focus on optimizing the computational efficiency of these models, potentially through more efficient embedding methods or streamlined symbolic reasoning processes.

Principled vs. Integrated Neuro-Symbolic Systems: There is a need to explore both principled and integrated approaches to neuro-symbolic AI. Principled systems maintain a clear separation between symbolic and neural components, while integrated systems combine these elements more fluidly. Research into the advantages and limitations of each approach can inform the development of hybrid AI systems that best meet clinical needs.

Enhancing Usability through Better Visualization and User Interfaces: Usability remains a critical challenge in deploying KG-based AI systems in clinical settings. Enhanced visualization techniques, as well as intuitive interfaces, can improve user experience and help clinicians interpret AI-generated recommendations more effectively. Additionally, understanding user needs, as highlighted by recent research, will guide the development of tools that align with clinical workflows and support decision-making processes. These future directions aim to refine the interpretability, usability, and efficiency of hybrid AI systems in healthcare, making them more practical in real-world applications. The integration of semantics within AI holds substantial promise, not only for advancing predictive accuracy but also for fostering trustworthy and human-centered AI solutions in medicine.

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